COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 5 : SLO – 1 | Signed magnitude representation
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Multiple Choice Questions (5)

• 1. d — Sign
• 2. b — 1101 (sign 1, magnitude 101)
• 3. c — Efficient hardware implementation (it is NOT a limitation — it is false; signed magnitude has complex hardware)
• 4. c — +2 (different signs: subtract magnitudes 5-3=2, sign of larger = +)
• 5. a — Positive zero (equal magnitudes, different signs → 0, conventionally +0)

Two-Mark Questions (2)

• 1. Signed magnitude is less preferred because it has two representations of zero (+0 and –0), and its arithmetic needs separate adder/subtractor hardware plus a magnitude comparator, making it slower and costlier than two's complement.
• 2. Rules of binary addition: 0+0=0; 0+1=1; 1+0=1; 1+1=0 with carry 1; 1+1+1=1 with carry 1.

Scenario-Based Long Answer (1)

• (-5) + (-3) in 4-bit signed magnitude: -5 = 1101, -3 = 1011. Same signs → add magnitudes: 101 + 011 = 1000 (8) and keep the common sign (negative).
• But the magnitude field is only 3 bits (max 7). 8 needs 4 bits (1000) → the sum overflows the magnitude field. Result: OVERFLOW — the true result -8 cannot be represented in 4-bit signed magnitude (range -7 to +7).

Match the Following (5)

• A. MSB in signed magnitude — 3. Sign bit
• B. 1 + 1 in binary — 4. 0 with carry 1
• C. Two's Complement — 1. Preferred representation today
• D. +0 and -0 — 2. Dual zero issue
• E. Overflow example — 5. +5 + +3 = +8 in 4-bit

Fill in the Blanks (5)

• 1. sign 2. 0 (with carry 1) 3. magnitudes 4. 0000 and 1000 5. two's complement

Scenario-Based Small Activity (1)

• +4 = 0100, -2 = 1010. Signs differ → compare magnitudes: 100 > 010. Subtract smaller from larger: 100 - 010 = 010 (2). Result takes the sign of the larger magnitude (positive).
• Final result: 0010 = +2.

True or False (5)

• 1. FALSE — It has two zeros (+0, -0). 2. TRUE
• 3. FALSE — 1+1 gives 0 with carry 1. 4. TRUE
• 5. FALSE — Overflow occurs when the magnitude sum exceeds the magnitude range (e.g. 5+3 in 4-bit).